Hyowon Kim

Ph.D. in Astrophysics

✉ *hwkim1011@gmail.com*
kimyo1@kasi.re.kr

Personal information

- Position : Postdoctoral researcher
- Affiliation : Korea Astronomy and Space science Institute (KASI), 776 Dae-Duk Daero, YuSung-gu, Daejeon 34055, Korea
- Office : Lee won cheal Hall, room 216
- homepage : https://kimyo1.github.io/hyowonkim/
- GitHub repository : https://github.com/kimyo1
- ORCID : 000-0003-4032-8572
- ADS library for publication : public-libraries/3oDpvOWnSH6GsH9UpU0bJg

Employment

March,2026–present **Korea Astronomy and Space science Institute**,
Galaxy evolution research center, Postdoctoral researcher.

June,2024–Feb,2026 **Universidad Tecnica Federico Santa Maria**,
Department of physics, Postdoctoral researcher.

Education

March,2017–Feb,2024 **University of Science and Technology**,
Department of Astronomy and Space science, Integrated course of Ph.D..

May,2010–Feb,2015 **Kyung pook National University**,
Department of Astronomy and Atmospheric science, Bachelor of Astronomy.

Research Interests

- Formation and evolution of Galaxy systems
- BCG-cluster co-evolution
- Environmental effect on galaxy and galaxy systems
- Effect from Large-scale structures on galaxy cluster
- Connection between High-z proto-cluster and LBG to current cluster and BCG
- Observational cosmology

Science collaboration

Chilean Cluster Galaxy Evolution Survey (CHANCES)

Nucleo Milenio de Galaxieas (MINGAL)

- Southern Photometric Local Universe Survey (SPLUS)
- KASI-Deep Rolling Imaging Fast Telescope (K-DRIFT)

Awards & Funds

2023 **ALMA ANID fund**, *Fund*.

- "New techniques for accurate catalogs of cluster dynamical states to study cluster evolution"
- 2024.∼ 2026.

2023 **UST Young Scientist fund**, *Fund*.
- "Study of the dynamical state of the galaxy cluster as a proxy of matter evolution of the universe"
- 2022.12.01.∼2023.11.30.

2022 **UST overseas program**, *Fund*.
- "Developing new recipes for measuring the dynamical state of galaxy clusters using simulations"
- 1st, Apr ∼ 26th, Jun, 2022
- Universidad Technica Frederico de Santa Maria, Santiago, Chile

Teaching, Supervision & mentoring

2025 **Special lecture for Graduate course, computational astrophysics**.
- USM, physics depart, FIS487, "Machine learning classification with Gaussian Mixture Model"

2025 **Co-supervising of student**.
- Patricio Olivares (USM, Master): relation between optical and X-ray dynamical state indicators in the CHANCES cluster catalog.

2024 **Examinator of midterm defense**.
- Tomas Fuetes (USM, Ph.D): Break BRD galaxy

Paper: First-author

3 **"New classification method for the dynamical state of galaxy clusters with a Gaussian mixture model"**.
- Hyowon Kim; Marco Canducci; Rory Smith; Peter Tino; Yara Jaffe; Hwang Ho Seong; Jihye Shin; Kyungwon Chun
- 2026, A&A, Accepted, (arxiv:2603.10319)

2 **"New observational recipes for measuring dynamical state of galaxy clusters"**.
- Hyowon Kim; Rory Smith; Jongwan Ko; Jongho Shinn; Kyungwon Chun; Jihye Shin; Jaewon Yoo
- 2024, ApJ, 970, 165K, Published

1 **"Is the Abell 2261 a fossil galaxy cluster in a transitional dynamical state?"**.
- Hyowon Kim;Jongwan Ko;Rory smith;Jae-Woo Kim;Ho-Seong Hwang;Hyunmi Song;Jihye Shin;Jaewon Yoo
- 2022, ApJ, 928, 170, published

Paper: Co-author

9 **"K-DRIFT Science Theme: New Theoretical Framework Using the Galaxy Replacement Technique for LSB studies"**.
- Chun, Kyungwon ; Shin, Jihye ; Smith, Rory ; Ko, Jongwan ; Yoo, jaewon ; Park, So-Myoung ; Byun, Woowon ; Chun, Sang-Hyun ; Hong, Sungryong ; Kim, Hyowon ; Kim, Jae-Woo ; Lee, Jaehyun ; Park, Hong Soo ; Rhee, Jinsu ; Seon, Kwang-Il ; Yoon, Yongmin
- 2026, JKAS, Accepted

8 **"K-DRIFT Science Theme: Galaxies in the Faint Universe"**.
- Byun, Woowon ; Yoon, Yongmin ; Ko, Jongwan ; Lee, Yun Hee ; Lee, Gain ; Hwang, Ho Seong ; Sabiu, Cristiano G. ; Seon, Kwang-il ; Chun, Kyungwon ; Shin, Jihye ; Rhee, Jinsu ; Kim, Jae-Woo ; Yoo, Jaewon ; Lee, Jaehyun ; Chun, Sang-Hyun ; Park, Hong Soo ; Yang, Soung-Chul ; Hong, Sungryong ; Shin, Jeehye ; Kim, Hyowon
- 2026, JKAS, Accepted

7 **"K-DRIFT Science Theme: Illuminating the Next Era of Galaxy Cluster Science"**.
- Yoo, Jaewon ; Chun, Kyungwon ; Ko, Jongwan ; Shin, Jihye ; Sabiu, Cristiano G. ; Lee, Jaehyun ; Seon, Kwang-il ; Kim, Jae-Woo ; Rhee, Jinsu ; Hong, Sungryong ; Byun, Woowon ; Kim, Hyowon ; Chun, Sang-Hyun ; Park, Hong Soo ; Yoon, Yongmin ; Shin, Jeehye
- 2025, JKAS, Accepted

6 **"The Role of Supercluster Filaments in Shaping Galaxy Clusters"**.
- Raúl Baier-Soto; Yara Jaffé; Alexis Finoguenov; Christopher P. Haines; Paola Merluzzi; Ulrike Kuchner; Rory Smith; Antonela Monachesi; Nicolas Tejos; Cristóbal Sifón; Diego Pallero; Ivan Lacerna; Maria Argudo-Fernández; Hugo Méndez-Hernández; Ciria Lima-Dias; Lawrence E. Bilton; Franco Piraino-Cerda; Ricardo Demarco; Alessia Moretti; D. Olave-Rojas; U. Rescigno; and CHANCES team
- 2025, A&A, Accepted

5 **"Tracing Dark Matter in the Central Regions of Galaxy Clusters Using Galaxies, Gas, and IntraclusterLight in TNG300: Connections to Cluster Dynamical State"**.
- Jaewon Yoo , Jihye Shin , Ho Seong Hwang , Cristiano G. Sabiu , **Hyowon Kim** , Jongwan Ko , and Jong Chul Lee
- 2025, ApJ, Published

4 **"Spatial Distribution of Intracluster Light versus Dark Matter in Horizon Run 5"**.
- Jaewon Yoo; Changbom Park; Cristiano G. Sabiu; Ankit Singh; Jongwan Ko; Jaehyun Lee; Christophe Pichon; M. James Jee; Brad K. Gibson; Owain Snaith; Juhan Kim; Jihye Shin; Yonghwi Kim;**Hyowon Kim**
- 2024, ApJ, Published

3 **"BCG alignment with the locations of cluster members and the large-scale structure out to 10_{R200}"**.
- Smith Rory;Ho Seong Hwang;Katarina Kraljic; Paula Calderon-Castillo;Thomas M. Jackson;Anna Pasquali;Jihye Shin;Jongwan Ko; Jaewon Yoo;**Hyowon Kim** ; Jae-woo Kim
- 2023, MNRAS, Published

2 **"How to quantify the similarity of 2D distributions:Comparison of spatial distribution of Intracluster light and Dark Matter"**.
- Jaewon Yoo;Jongwan Ko;Jihye Shin;Cristiano G. Sabiu;Kyungwon Chun;Ho Seong Hwang;Juhan Kim;M. James Jee;**Hyowon Kim**;Rory Smith
- 2022,ApJS,261,28, Published

1 **"Intracluster light properties in a fossil cluster at z = 0.47"**.
- Jaewon Yoo;Jongwan Ko;Jae-Woo Kim;**Hyowon Kim**
- 2021,MNRAS,508,2634Y, Published

Conference and Talk

recent 3 years **3 invited seminar talk, 5 conference talks and two contribution talks** , *Korea, Chile, Japan, US, Brazil*.
- "New classification method of dynamical state of galaxy clusters using Gaussian Mixture Model"
- Oral talk

Observing Experience

2026A **Testing the formation hypothesis of the fossil system by estimating the stellar assembly history of fossil BCGs** .
- P.I. Jaewon Yoo (Proposal co-author)
- Gemini South longslit spectroscopy observation
- Proposal writing, Data reduction

2025FT **Probing the Co-Evolution of Brightest Cluster Galaxies and Intracluster Light with Gemini GMOS: A Spectroscopic Follow-up of LSST ComCam Data**.
- P.I. Jaewon Yoo (Proposal co-author)
- Gemini-South MOS spectroscopy observation
- Target selection, Phase II design

2019A&B **Probing the galaxy populations of merging galaxy clusters from filaments to the core**.
- P.I. Kyle finner
- MMT Hectospec Spectroscopy observation
- Proposal co-author

2018A **Constraining the ICL formation mechanism using fossil clusters at redshift 0.45**.
- P.I. Jaewon Yoo
- Gemini-North Optical observation
- Proposal co-author

2017B **Tracing the origin of the intracluster light in massive galaxy clusters at z>1 : a case of MOO J1014+0084**.
- P.I. Jongwan Ko
- Gemini-South slit spectroscopy observation
- Phase II design, On-site observation (2nights)

2017A **Addressing The Evolutionary History of Galaxy Clusters through Member Galaxies**.
- P.I. Jae-Woo Kim
- MMT Hectospec Spectroscopy observation
- Data reduction

Attended Courses and Schools

2023 K-GMT summer school,*KyungJu, Korea*

2020,2021 KIAS cosmology summer school,*ZOOM*

2018 Summer School in Statistics for Astronomers XIV,*Pennsylvania University, USA*

2017 Gravitational wave school,*Sejong, Korea*

Other activities

2021-2024 Student committee of preventing sexual harassment in KASI

2020-2021 Korea astronomy encyclopedia assistant (public outreach)

2019 Group student meeting organizer

2018 UST Student representative

2018 UST Journal club organizer

2015-2016 KNU University assistant officer of major

Experience of Data

Optical SDSS, HSC, S-PLUS

Spec HeCS, GMOS MOS & longslit

Sim N-cluster run simulation, Illustris-TNG

Skills

programming IDL, Python, HTML/CSS, JavaScript

Utilities Topcat, ds9, IRAF, specpro, HSRED, SExtractor, SCAMP, SWARP, pPXF, Gemini Dragons

Communication Korean (native), English (TOFLE: 81 (2021.11), fluent), Spanish (beginner)

Development of program bcdc (Bayesian classifier for dynamical states of clusters of galaxies): https://github.com/kimyo1/bcdc

References

Jongwan Ko, *Ph.D Supervisor*.
- Affiliation :
- 1) Korea Astronomy and Space science Institute, Staff scientist
- 2) University of Science and Technology, Professor
- E-mail : jwko@kasi.re.kr

Rory Smith, *Professor*.
- Affiliation : Universidad Technica Frederico de Santa Maria, Santiago, Chile
- Associate Professor
- E-mail : rorysmith274@gmail.com
- webpage: https://rorysmith274.wixsite.com/mysite

Yara Jaffe, *Professor*.
- Affiliation : Universidad Technica Frederico de Santa Maria, Valparaiso, Chile
- Associate Professor
- E-mail : yara.jaffe@usm.cl
- ORCID: https://orcid.org/0000-0003-2150-1130